

Adaptation of Generalist Robot Policies with Minimal Data

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Abstract: A central goal of robot learning is to move beyond task-specific human data collection toward robots that improve through autonomous interaction. Yet fully autonomous learning remains difficult with current policies: sparse rewards and weak zero-shot exploration make it unlikely that a robot will discover successful behavior from scratch. We study **minimal-data adaptation**, a regime in which a pretrained robot policy must learn a new task from as little as one demonstration followed by autonomous online interaction. This setting minimizes task-specific supervision while testing whether pretrained policies provide enough structure to make reinforcement learning tractable. We present **MIDAS**, an offline-to-online RL framework that first anchors a pretrained VLA to the target task with few-demo behavior cloning, then improves it through value-based online residual RL. Across LIBERO and RoboCasa, MIDAS recovers strong task performance from as little as one demonstration, substantially outperforming baselines and generalizing beyond demonstrated conditions. To the best of our knowledge, this is the first demonstration of reliable robot policy adaptation from a single task demonstration. We further validate MIDAS on bimanual YAM hardware, where it learns effective behaviors with roughly six hours of autonomous interaction.

1 Introduction

A long-standing goal in robotics is to deploy robots that improve through their own experience: a robot enters a new environment, attempts a task, and refines its behavior until it succeeds reliably, without further human supervision. While autonomous adaptation faces challenges at several levels, in this work, we focus on the algorithmic barriers to fast robot adaptation to a new task. Prior work has studied autonomous adaptation largely with specialist, single-task policies [1, 2, 3], often requiring complex task-specific methodology. The prevalence of generalist robot policies [4, 5, 6, 7, 8, 9, 10], pretrained on Internet data makes it natural to revisit this question: can a broad pretrained prior make online adaptation possible from minimal task-specific supervision?

To answer this question, we first note that autonomous adaptation requires three ingredients. The policy must represent high-dimensional observations in a form useful for control and value learning, produce behavior that reaches reward-bearing regions during deployment, and use this experience to improve efficiently. Recent generalist robot policies, including vision-language-action (VLA) models, make progress on the first two ingredients [4, 5, 6, 7, 8, 10]: their representations provide useful features for control, and their action heads encode behavioral priors learned from diverse manipulation data. However, these policies are still not reliable zero-shot learners in new environments. They may fail to follow the task instruction, miss sparse reward entirely, or visit only a narrow set of useful states, leaving online learning with too little signal to bootstrap improvement.

We study this gap by proposing the *minimal-data adaptation (MDA)* setting, in which a pretrained policy must learn to improve autonomously from only minimal task-specific supervision, such as a single successful demonstration. Prior work has developed many of the tools needed for this setting, but typically assumes either that the initial policy already has substantial task compe-

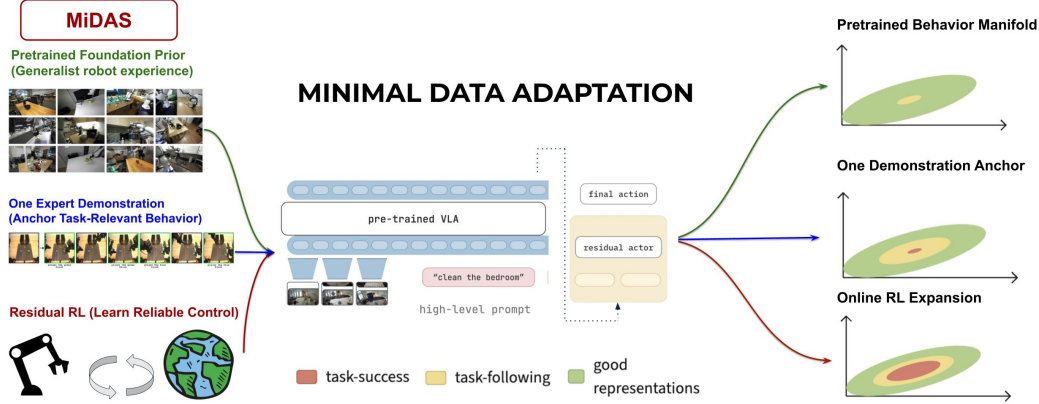


Figure 1: Overview of minimal-data adaptation . A single demonstration is sparse and underdetermined on its own, but pretrained VLA representations turn it into a useful anchor for adaptation. Few-demo behavior cloning narrows the search space to a task-relevant region around the demonstration, and online residual RL then expands local success by learning value-guided corrections on top of the warm-started base policy.

tence [11, 12, 13] or that tens to hundreds of demonstrations are available [14, 15, 13, 16]. Few-shot imitation learning methods suggest that learning from one or two demonstrations is possible in some settings [17, 18, 19], but most available VLA policies do not naturally exhibit reliable few-shot improvement (related work discussed in Appendix B). MDA targets the narrow regime between zero-shot deployment and standard task-specific finetuning: the pretrained policy is not reliable enough to deploy directly, but may be close enough that one demonstration can make online learning tractable.

Our **central finding** is that minimal-data adaptation is already tractable for modern generalist robot policies. Even just a single successful demonstration can make autonomous adaptation possible, not because one demonstration provides enough coverage for imitation learning, but because adaptation succeeds through a division of labor across different pieces. Pretraining provides reusable features and broad action priors, but these priors are not calibrated for reliable performance on the target task. The demonstration anchors the policy to the task and makes task-relevant behavior reachable, but does not provide reliable control. Autonomous interaction supplies the remaining corrections by learning from the robot’s successes and failures. We instantiate this principle in **MIDAS**: an offline-to-online recipe that first finetunes a pretrained VLA by imitation on K demonstrations to obtain a task-anchored policy and then freezes the VLA but trains a lightweight residual actor online on top of its representations using value-based RL [20] on data collected via autonomous interaction.

We make three contributions in this paper. **First**, we introduce *minimal-data adaptation*, a regime where a pretrained robot policy is not reliable zero-shot, but may be close enough that one or a few demonstrations make online learning tractable. **Second**, we present **MIDAS**, a value-based deep RL method that combines few-demo behavior cloning with frozen-backbone residual RL and stabilizing choices for sparse-reward adaptation. Across LIBERO-Long and RoboCasa, **MIDAS** recovers strong performance from as little as one demonstration, outperforms baselines, generalizes beyond demonstrated conditions, and transfers to a real bimanual YAM platform with roughly 6 hours of autonomous interaction. **Third**, we use this regime to explain why adaptation works: behavior cloning recovers task-level behavior but not reliable control, pretrained VLA representations enable sample-efficient value learning, and online residual RL supplies corrective actions beyond the few-demo policy’s effective support. Together, these results show that modern pretrained robot policies can use a single demonstration to scaffold autonomous adaptation, while clarifying what how pretraining, task specification, and online experience determine adaptation ability.

2 Problem Setting: Minimal Data Adaptation

We now define the minimal-data adaptation (MDA) setting and motivate why it is a useful regime to study. Given a pretrained vision-language-action policy π_{base} , K successful demonstrations $\mathcal{D}_{\text{demo}}^K$ collected in a target environment (denoted by $\mathcal{M}_{\text{target}}$), and access to autonomous interaction with

74 $\mathcal{M}_{\text{target}}$, minimal-data adaptation asks whether we can learn an adapted policy π_{θ}^K that achieves
 75 high return in $\mathcal{M}_{\text{target}}$ using only this limited supervision. The goal is not to solve the task from
 76 demonstrations alone. Instead, the demonstrations serve as a minimal scaffold that makes sub-
 77 sequent autonomous learning tractable. The case $K = 1$ is the sharpest version of this setting:
 78 learning on a single demonstration is often insufficient to learn previously unseen tasks, but may
 79 still expose enough task-relevant behavior for interaction-driven improvement to begin.

80 MDA targets the narrow regime where autonomous learning first becomes tractable. With no task-
 81 specific data, current pretrained policies often fail to reach task-relevant behavior often enough for
 82 sparse-reward interaction to improve them. A small number of demonstrations can make the problem
 83 just structured enough: they expose the intended behavior and bring the policy into the vicinity of
 84 success, while still leaving robustness and distribution coverage to be learned through autonomous
 85 interaction. This setting therefore tests whether pretrained robot policies have reached a regime
 86 where a few demonstrations can specify the task while online interaction supplies the missing robust-
 87 ness. It also separates the roles of the available sources of supervision: the structure inherited from
 88 pretraining, the task information provided by demonstrations, and the corrections learned through
 89 autonomous interaction. Our approach MIDAS uses value-based residual RL to perform this online
 90 refinement; we briefly review these primitives before describing the full adaptation recipe.

91 3 Preliminaries

92 We formalize MDA in the standard reinforcement learning setting. The goal is to learn a policy
 93 $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ that maximizes the discounted return $J(\pi; \mathcal{M}) = \mathbb{E}^{\mathcal{M}, \pi}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$, where
 94 $s_0 \sim \rho$, $a_t \sim \pi(\cdot | s_t)$, and $s_{t+1} \sim P(\cdot | s_t, a_t)$. We focus on sparse binary rewards, where $r_t = 0$
 95 if the task is completed within the time limit and -1 otherwise. A central object in our method is the
 96 action-value function $Q^{\pi}(s, \mathbf{a})$, which estimates the expected discounted return from taking action
 97 $\mathbf{a}$ in state s and following π thereafter. In our robotic setting, states include high-dimensional visual
 98 and proprioceptive observations, and actions are temporally extended chunks [21]: each $\mathbf{a}_t$ denotes
 99 a sequence of low-level controls executed open-loop. See Appendix A for details.

100 **Value-Based RL.** An approach to value-based RL maintains a replay buffer $\mathcal{B}$ of previously col-
 101 lected transitions and trains a critic using temporal-difference (TD) learning. Given a transition
 102 $(s_t, \mathbf{a}_t, r_t, s_{t+1}) \sim \mathcal{B}$, the TD target and loss are

$$y_t = r_t + \gamma Q_{\bar{\psi}}(s_{t+1}, \mathbf{a}'), \quad \mathcal{L}_{\text{TD}}(\psi) = \mathbb{E}_{\mathcal{B}} \left[(Q_{\psi}(s_t, \mathbf{a}_t) - y_t)^2 \right], \quad (3.1)$$

103 where $Q_{\bar{\psi}}$ is a target critic and $\mathbf{a}'$ is the next action. The policy π_{θ} is then updated toward actions
 104 that receive high value under the critic. [22, 23]

105 **Residual RL.** Residual RL adapts a frozen base policy by learning a lightweight policy module that
 106 modifies the base policy’s proposed action [24, 25, 26, 27, 28]. Given a base policy π_{θ} , the residual
 107 policy $\pi_{\theta}^{\text{res}}$ conditions on both the state and an action proposed by the base policy, and outputs the
 108 action executed in the environment. This residual update can be parameterized in several ways, such
 109 as predicting an additive offset in action space or directly predicting the modified action. In this
 110 work, we adopt the latter parameterization as discussed in Section 4.

111 **Policy-Agnostic RL.** Policy-agnostic RL (PA-RL) [20] is an online improvement method that ap-
 112 plies across policy classes. It first improves sampled actions against the learned Q-function, then
 113 distills the optimized actions back into the parametric policy using the appropriate supervised ob-
 114 jective, such as flow matching for flow policies or likelihood for discrete-action policies. We use
 115 PA-RL for this policy-agnostic structure, extending it to action-chunked residual policies in Ap-
 116 pendix E.1. For each replay state s_t , we sample candidate action chunks, refine them by optimizing
 117 directly against Q_{ψ} to obtain $\mathbf{a}_t^*$, and distill the optimized chunk back into the residual policy.

118 4 MIDAS : A Simple Recipe for Minimal Data Adaptation

119 In this section, we introduce **Minimal-Data Adaptation Strategy** (MIDAS), a two-stage recipe for
 120 adapting a pretrained policy to a new task from only a handful of demonstrations. What can such

minimal supervision realistically provide? As shown in Section 5 and depicted in Figure 1, fine-tuning a pre-trained model with K initial demonstrations seeds coarse task-following behavior, but does not ensure reliable completion. This perspective leads to a simple two-stage recipe. **Stage I** fine-tunes π_{base} on the K demonstrations with behavior cloning, producing a policy π_{base}^K that can attempt the correct task but remains unreliable. **Stage II** freezes π_{base}^K and trains a lightweight residual actor-critic on top of its representations using sparse-reward online RL. The resulting policy, π_{RL}^K , provides the fine-grained control corrections needed for a high-success rate policy.

Stage I: Behavior cloning on minimal demonstrations. Stage I adapts the pretrained policy π_{base} to the downstream task using the K demonstrations in $\mathcal{D}_{\text{demo}}^K$. We adapt the VLM backbone using a low-rank adapter (LoRA [29]) for compute efficiency, and fine-tune the complete action head using the standard flow matching objective. Implementation details are deferred to Appendix E.2. As described in Section 5, the resulting π_{base}^K coarsely attempts the target task, but does not complete it reliably.

Stage II: Online improvement via value-based residual RL. Stage II improves π_{base}^K through autonomous interaction using value-based RL. Because Stage I produces a policy that can roughly complete the tasks, Stage II need only provide small corrections to overcome control failures. Therefore, to anchor RL updates around π_{base}^K , we freeze the pretrained policy backbone and train only a lightweight adaptation module on top of it. Following common practice in prior work [24, 25, 26, 27, 28], we instantiate this module as a policy that predicts a correction to the action proposed by π_{base}^K . At each state s_t , the base policy proposes an action chunk $\mathbf{a}_t^{\text{base}}$, and the residual policy $\pi_{\theta}^{\text{res}}$ proposes an edit conditioned on the state and the chunk itself. We parameterize the residual policy as follows¹

$$\pi_{\theta}^{\text{res}}(\cdot | s_t, \mathbf{a}_t^{\text{base}}) = \tanh(\mathcal{N}(\mu_{\theta}(s_t, \mathbf{a}_t^{\text{base}}), \sigma_{\theta}^2(s_t, \mathbf{a}_t^{\text{base}}))). \quad (4.1)$$

Conditioning the actor on $\mathbf{a}_t^{\text{base}}$ exposes it directly to the action mode selected by π_{base}^K , so online RL refines a task-relevant VLA proposal rather than learning an action distribution from scratch. At the same time, predicting $\mathbf{a}_t^{\text{exec}}$ directly allows the policy to shift this proposal toward higher-value regions instead of being restricted to small fixed residual corrections around $\mathbf{a}_t^{\text{base}}$ that further limit the policy. Then, we evaluate this action by a critic ensemble $\{Q_{\psi_j}(s_t, \mathbf{a}_t^{\text{exec}})\}_{j=1}^N$ trained with temporal difference learning in Eq. (3.1). We discuss implementation details for this residual parameterization in Appendix C.

Overcoming the challenges introduced by minimal demonstrations. Given minimal demonstrations, few expert trajectories exist for initializing the learned critic and residual-actor. Moreover, autonomous rollouts from π_{base}^K succeed only rarely, for example 2-5% of the time. Lastly, minimal demonstrations need not ensure that π_{base}^K contains optimal action in its support (Section 5). We now introduce key design choices to overcome these respective challenges.

1) Offline warmup: residual actor initialization and critic calibration from few demonstrations. A randomly initialized residual actor would produce actions unrelated to the base policy, making early exploration unstable. We therefore construct a “warmup” buffer $\mathcal{B}_{\text{warm}}$ from the K demonstrations in $\mathcal{D}_{\text{demo}}^K$ together with a small number of autonomous rollouts from π_{base}^K that may or may not be successful. For each state in this buffer, we query the frozen base policy to obtain $\mathbf{a}^{\text{base}} \sim \pi_{\text{base}}^K(\cdot | s)$, and train the residual actor to reproduce this base proposal:

$$\mathcal{L}_{\text{warm}}(\theta) = \mathbb{E}_{(s, \mathbf{a}^{\text{base}}) \sim \mathcal{B}_{\text{warm}}} \left[\left\| \mu_{\theta}(s, \mathbf{a}^{\text{base}}) - \mathbf{a}^{\text{base}} \right\|_2^2 \right]. \quad (4.2)$$

This initialization makes the residual actor approximately preserve the behavior of π_{base}^K : Online RL also requires a critic that can guide improvement from the start. While prior work can pretrain such critics from offline data [14, 15, 30, 31], our setting has no additional offline data beyond the few demonstrations and autonomous rollouts from π_{base}^K . We therefore draw inspiration from Zhou et al. [30] and train the critic on the warmup buffer $\mathcal{B}_{\text{warm}}$ using TD learning (Eq. (3.1)). Importantly, this critic is trained on the state-action distribution induced by π_{base}^K , making it calibrated to the regions where online improvement begins and limiting forgetting.

¹Unlike typical residual formulations, (4.1) directly predicts the edited action $\mathbf{a}_t^{\text{exec}} \sim \pi_{\theta}^{\text{res}}(\cdot | s_t, \mathbf{a}_t^{\text{base}})$.

168 **2) Success Balancing: overcoming limited initial policy performance.** Since the base policy may
 169 only succeed 2-5% in some cases, we employ *success balancing* to amplify the sparse reward signal
 170 during both the warmup and online improvement phases. To do so, we maintain a standard replay
 171 buffer $\mathcal{B}$ containing all collected trajectories and a success buffer $\mathcal{B}_{\text{succ}}$ containing only successful
 172 trajectories, both initialized with the same K demonstrations. Critic minibatches are sampled from
 173 a mixture of these buffers, oversampling $\mathcal{B}_{\text{succ}}$ to prevent the value function from being dominated
 174 by failed rollouts and to guide policy improvement toward reward-bearing regions. Further details
 175 are present in Appendix E.3.

176 **3) Residual policy training with PA-RL [20].** Lastly, Section 5 reveals the need for actions that go
 177 beyond the support of the base policy. While many value-based RL algorithms can provide this, we
 178 opt for “policy-agnostic” PA-RL approach, which we apply to the residual actor. To do so, we sample
 179 a *single* proposal action $\mathbf{a}_t^{\text{base}}$ from the base policy, then apply PA-RL style Best-of-N distillation
 180 objective with an action gradient to a batch of N actions $\mathbf{a}_t^{(i)} \sim \pi_{\theta}^{\text{res}}(\cdot | \mathbf{s}_t, \mathbf{a}_t^{\text{base}})$ sampled from the
 181 residual. See Appendix E.1 for more details. Importantly, sampling requires *only* a single expensive
 182 forward pass through π_{base}^K and no backward passes since the base is frozen.

183 5 What Enables MIDAS to Learn Effectively?

184 Previously, we motivated MIDAS with the premise that limited demonstrations can seed behavior
 185 close enough to success to seed online improvement. In this section, we present detailed empirical
 186 evidence identifying how the three sources of supervision that enable MIDAS to achieve this: First,
 187 the few-demo BC finetuning of the pre-trained VLA provides actions which follow task instructions
 188 coarsely, reducing the burden of exploration. Second, the pretrained VLA provides visual repre-
 189 sentations, enabling efficient learning without direct state access. Finally, online reinforcement fills
 190 in the gaps in precise-control, which necessarily requires value-guided online updates to finetuning
 191 actions beyond what π_{base}^K can reliably produce on its own. Additional ablations of MIDAS are
 192 deferred to Appendix F and experiments to Appendix I.

193 **Experimental setup.** We use $\pi_{0.5}$ [7] as the pretrained base policy π_{base} and run MIDAS with
 194 $K = 1$ demonstration only on two simulation benchmarks: **a)** LIBERO-Long [32], which studies
 195 long-horizon language-conditioned manipulation, and **b)** RoboCasa-365 [33], which studies long-
 196 horizon household manipulation in diverse kitchen scenes (Details in Appendix D.1). By ablating
 197 different components of our approach, we aim to identify the core mechanisms that make MIDAS
 198 work well.

199 1) Pretraining and task-specific demonstrations enable coarse behavior recovery.

200 Before online adaptation, we ask whether $K=1$ can re-
 201 cover task-anchored behavior. Figure 2 compares π_{base}^1
 202 to three approaches: **a)** a flow-matching policy trained on
 203 pretrained ResNet features, which tests whether generic
 204 visual pretraining is sufficient; **b)** a flow-matching pol-
 205 icy trained on privileged state inputs, which tests whether
 206 near-optimal state information alone enables recovery of
 207 task-anchored behavior; and **c)** the zero-shot pretrained
 208 policy π_{base} , which tests whether pretraining alone iden-
 209 tifies the task. All three perform poorly: the ResNet flow-
 210 policy achieves non-zero success on only 3/10 LIBERO-
 211 Long tasks, while the privileged-state policy and π_{base}
 212 obtain 0% success. Together, these failures suggest that
 213 MDA requires both task grounding from the demonstra-
 214 tion and behavior-aware representations from pretraining; privileged state inputs alone do not re-
 215 cover this behavior, suggesting that the benefit of pretraining goes beyond improved state represen-
 216 tation.

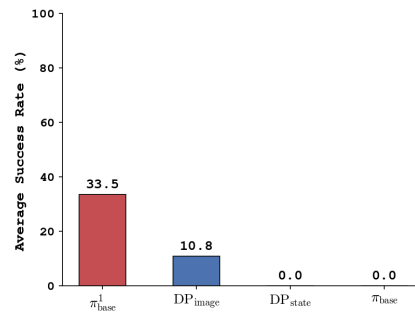


Figure 2: Performance of different task representations for behavior cloning

217 Qualitative rollouts in Fig. 3 show what π_{base}^1 recovers despite its low success. The policy often
 218 approaches the correct object, attempts the correct subtask sequence, and follows a trajectory re-

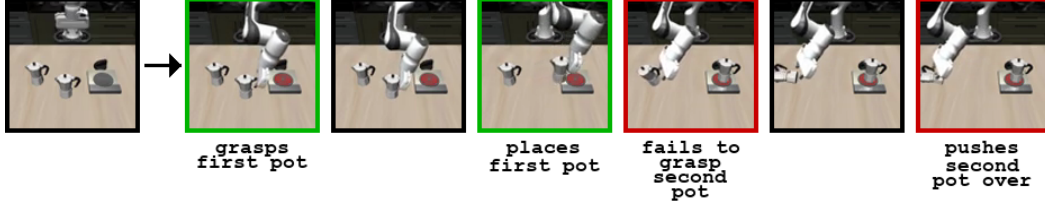


Figure 3: Few-demo BC recovers task-level behavior, but not fine-grained control. Rollouts of π_{base}^K at $K=1$ on *Both Moka Pots* $\rightarrow$ *Stove*. The policy reaches the right objects but fails to grasp the second pot under orientation changes.

219 sembling the demonstration. Its failures are primarily in reliable execution: contact, alignment,
 220 grasping, and recovery remain brittle. Thus, pretraining and one demonstration learn coarse task
 221 behavior, but online adaptation is needed to solve the control problem.

2) Pretrained representations simplify on-line adaptation and substantially improve sample-efficiency.

224 The preceding results show that large-scale pretraining, together with minimal demonstrations, anchors the search for task-relevant behavior. We now find that pre-trained VLA representations also simplify the subsequent online learning problem, making
 229 sparse-reward residual RL substantially more sample-efficient. Figure 4 isolates this effect by vary-
 230 ing the representation used by the residual learner. With a ResNet fine-tuned from scratch, MiDAS
 231 fails to converge at $K=1$, indicating that sparse interaction is insufficient to learn both task-relevant
 232 visual features and fine-grained action priors for control. Generic visual pretraining helps, but frozen
 233 DINO features still require substantially more interaction, suggesting that visual invariances alone
 234 are not enough to serve as good pre-trained features. In contrast, features from the pretrained $\pi_{0.5}$
 235 backbone enable rapid improvement, suggesting that VLA pretraining yields representations already
 236 aligned with robot control. Finally, π_{base} , π_{base}^K , and π_{base}^{50} features perform similarly, implying that
 237 the gain in sample-efficiency can primarily be attributed to pretrained representations.

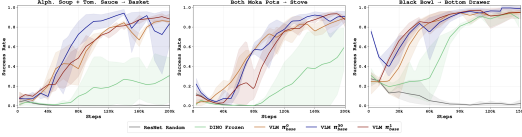


Figure 4: Performance of different representations in online adaptation

239 **3) Value-based RL learns fine-grained control actions beyond the base policy’s prior.** The
 240 previous analysis showed that π_{base}^K often produces task-relevant behavior, but solves the task in-
 241 consistently. We ask: do the right actions already lie within its support but are not reliably selected,
 242 or do they fall outside its effective support?

243 We test this by comparing
 244 MiDAS to two representative
 245 alternatives: *Filtered BC*, which
 246 finetunes π_{base}^K on successful
 247 online rollouts and sharpens
 248 modes the policy already
 249 reaches; and *Diffusion-Steering*
 250 *RL* (DSRL) [11], which steers
 251 within the action distribution of
 252 π_{base}^K by modifying its initial
 253 noise. In contrast, MiDAS
 254 applies value-guided residual
 255 corrections, allowing executed
 256 actions to move beyond high-
 257 probability samples from the
 258 base policy.

Table 1: One-demo adaptation results measured over 3 seeds. Success rates (%) are mean $\pm$ std.

Task	1 Demo			
	BC	DSRL	Filt. BC	MiDAS
<i>LIBERO-Long</i>				
Alph. Soup + Cr. Cheese $\rightarrow$ Basket	12.7 $\pm$ 1.2	16.0 $\pm$ 6.9	14.0 $\pm$ 2.0	98.7$\pm$1.2
Alph. Soup + Tom. Sauce $\rightarrow$ Basket	4.0 $\pm$ 3.5	3.3 $\pm$ 1.2	2.0 $\pm$ 0.0	82.0$\pm$5.3
Black Bowl $\rightarrow$ Bottom Drawer	22.7 $\pm$ 6.4	26.7 $\pm$ 1.2	28.7 $\pm$ 10.3	96.7$\pm$2.3
Book $\rightarrow$ Caddy	37.3 $\pm$ 7.0	30.7 $\pm$ 12.2	28.0 $\pm$ 2.0	92.7$\pm$5.0
Both Moka Pots $\rightarrow$ Stove	13.3 $\pm$ 3.1	20.7 $\pm$ 2.3	19.3 $\pm$ 7.0	90.7$\pm$2.3
Cr. Cheese + Butter $\rightarrow$ Basket	34.0 $\pm$ 9.2	39.3 $\pm$ 1.2	68.0 $\pm$ 2.0	100.0$\pm$0.0
Moka Pot $\rightarrow$ Stove	84.0 $\pm$ 0.0	92.7 $\pm$ 3.1	85.3 $\pm$ 8.3	94.0$\pm$0.0
White Mug + Choc. Pudding	33.3 $\pm$ 11.7	33.3 $\pm$ 10.3	39.3 $\pm$ 2.3	85.3$\pm$4.2
White Mug + Plates	30.0 $\pm$ 2.0	23.3 $\pm$ 3.1	34.7 $\pm$ 6.4	76.0$\pm$2.0
Yellow-White Mug $\rightarrow$ Microwave	64.0 $\pm$ 2.0	47.3 $\pm$ 8.3	68.0 $\pm$ 7.2	84.7$\pm$4.2
Average	33.5 $\pm$ 1.8	33.3 $\pm$ 2.0	38.7 $\pm$ 4.8	90.1$\pm$0.4
<i>RoboCasa</i>				
Banana: Fridge Drawer $\rightarrow$ Shelf	13.3 $\pm$ 2.5	13.9 $\pm$ 0.2	19.3 $\pm$ 3.1	95.2$\pm$2.8
Hot Dog: Counter $\rightarrow$ Cabinet	29.3 $\pm$ 0.9	17.6 $\pm$ 8.4	36.0 $\pm$ 24.0	87.1$\pm$8.6
Mug $\rightarrow$ Coffee Machine + Start	24.0 $\pm$ 7.1	24.7 $\pm$ 3.7	46.7 $\pm$ 6.2	81.9$\pm$5.7
Cup + Bowl $\rightarrow$ Dishwasher + Close	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Average	22.2 $\pm$ 3.5	18.7 $\pm$ 4.1	34.0 $\pm$ 10.6	88.1$\pm$5.7

259 Empirically, Filtered BC and
 260 DSRL improve only marginally over π_{base}^K , while MiDAS achieves substantially higher success
 261 **Table 1.** Fig. 5 makes this distinction clear, at a critical state in a LIBERO-Long rollout: actions

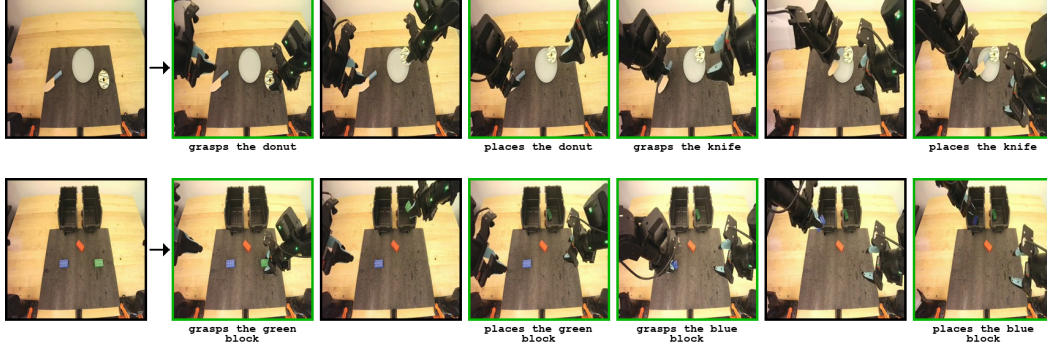


Figure 6: Qualitative rollout filmstrips for two real-world tasks

from Filtered BC and DSRL remain concentrated near samples from π_{base}^K and fail, while MiDAS selects an action in a disjoint region of the action space and succeeds. The fine-grained control missing from π_{base}^K in this case does not lie within the base policy’s effective action prior; sharpening or steering within that prior cannot reach it. Closing the control gap, therefore, requires online adaptation that can expand the action distribution beyond the effective support of π_{base}^K .

4) MiDAS transfers to the real world

We finally evaluate whether minimal-data adaptation is tractable in the real world using a bi-manual YAM platform. We consider two pick-and-place tasks: (T1) placing a green block in the right container and a blue block in the left container, and (T2) placing a knife and a donut on a plate (Figure 6). In both tasks, we use $\pi_{0.5}$ [7] as π_{base} and apply the same MiDAS protocol with a single demonstration ($K=1$); implementation details are deferred to Appendix J. On T1, the one-demo policy π_{base}^1 already reaches 40% success, which we hypothesize reflects overlap with the pretraining distribution of $\pi_{0.5}$. Online RL further improves success to 67% and produces qualitatively new corrective behaviors not observed in base-policy rollouts (Appendix J). On the harder T2 task, where the warm-start policy is substantially weaker, online adaptation improves success from 27% to 80% after 5-6 hours of autonomous interaction. These results show that MiDAS can be effective on real hardware, even when the one demo warm-started policy is not very reliable.

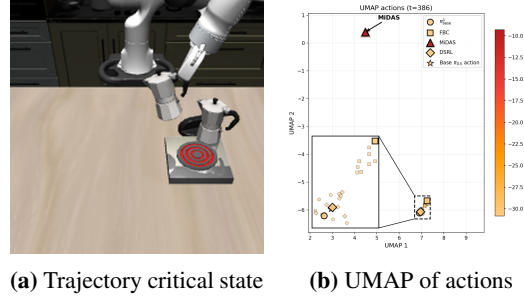


Figure 5: MiDAS reaches actions outside the effective support of π_{base}^K . UMAP of action chunks at a critical step of a LIBERO-Long rollout. Filtered BC and DSRL remain concentrated near samples from π_{base}^K and fail; MiDAS selects an action in a disjoint region with higher Q^* and succeeds.

6 Evaluating the Generalization and Robustness of MiDAS

Above, we exposed how the pre-trained policy, demonstrations, and online-improvement enable MiDAS to improve performance from the few initial conditions shown in the demonstrations. Here, we ask: can MiDAS generalize beyond the narrow support of initial states, and if so, how?

To test generalization, we consider two categories of shifts: (a) **observation shifts**, which change task-irrelevant inputs such as object appearance (color, texture) or language phrasing (instruction paraphrases), and (b) **state shifts**, which change dynamically relevant quantities such as starting positions, object shapes, or object instances. We ask: how does MiDAS generalize (1) to observation shifts in vision and language? (2) to shifts in state shifts in position, shape, and object identity? and (3): can autonomous learning alone improve generalization to larger state shifts? As in Section 5, we evaluate on the 10 long-horizon tasks in LIBERO-Long. For visual, shape, and object-instance shifts, we use the perturbation definitions from LIBERO-PRO [34]. For position shifts, we perturb target objects at three levels: low (5 cm / 0.5 rad), medium (10 cm / 0.75 rad), and high (20 cm / 1.0 rad). (Further details in D.2)

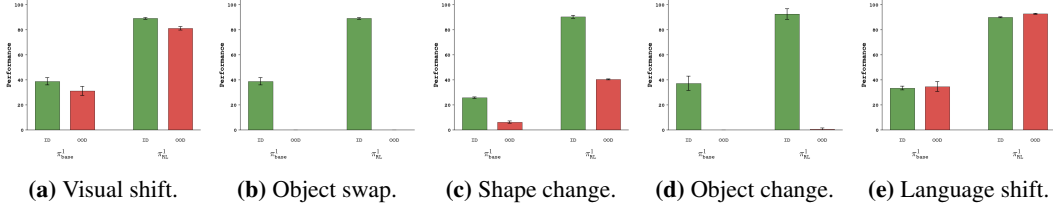


Figure 7: Performance of BC and online RL under observation and state shifts. **(a, e)** Observation shifts: pretrained representations preserve performance under visual and language changes. **(b–d)** State shifts: online RL improves robustness to moderate shifts but performance degrades under larger shifts.

(Q1) Observation-level robustness is inherited from the frozen vision-language backbone. We first test whether minimal-data adaptation preserves robustness to semantic input changes, rather than merely fitting the visual and language conditions seen in the demonstrations. Under visual shifts such as color and texture changes (Fig. 7a), π_{base}^K ($K = 1$) retains most of its in-distribution performance, and the adapted policy π_{RL}^1 shows a similar retention pattern despite using only a lightweight residual actor. The same holds under language paraphrases (Fig. 7e). Since both policies share the same frozen VLM backbone, and the RL module adds little perceptual or linguistic capacity, this robustness is best explained by invariances already present in the pretrained representation. Thus, online RL improves control without sacrificing observation-level generalization.

(Q2) Generalization to state shifts depends on whether new behavior is required. We next evaluate shifts that require the policy to change its actions. Figures 7c, 7b, and 7d show that π_{RL}^1 improves over π_{base}^1 under some state shifts, especially shape changes where the required grasp is similar. This makes sense: the policy must execute the same behavior in a different part of the state space. However, performance under both drops sharply under object swaps and object changes: changes in object affordance or placement can require distinct manipulation strategies, which are difficult to infer from one demonstration and online interaction around it. (Further details in D.3).

(Q3) Broader coverage, either from demonstrations or curriculum, expands the generalization boundary. Lastly, we ask: can autonomous adaptation ensure robustness to large perturbation shifts that are outside the support of the reset distribution, provided that they do not require fundamentally new behavior (in view of Q2). In the Appendix H, we show that naively increasing the breadth of demonstration can improve robustness. In Appendix G, we describe a curriculum that gradually increases the width of the distribution, and achieves similar robustness to a policy trained on 50 demonstrations.

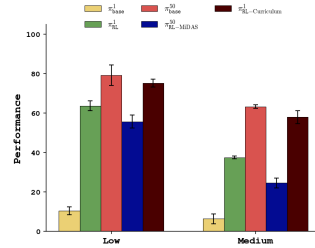


Figure 8: Curriculum closes the gap. Progressively widening the reset distribution during online RL recovers most of the position-generalization gap with π_{base}^{50} without additional demos.

7 Limitations

The results in this paper should be interpreted within two limitations. First, the present instantiation remains limited on longer-horizon tasks, where sparse-reward value learning must assign credit through long sequences of actions proposed by a frozen prior. Scaling MDA may therefore require stronger value priors, such as pretrained critics or value decompositions that expose intermediate progress signals. Second, Stage 2 remains reset-dependent and interaction-intensive. This is manageable in simulation, but becomes a central constraint on hardware, where failures, resets, and safety limit autonomous practice without an operator in the loop. Reducing this dependence will require safer exploration, better failure detection, learned resets, or lightweight intervention mechanisms that preserve the low-demonstration setting. We discuss directions for future work in Appendix K.

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448 A Action chunking

449 We apply action chunking [21], where sequences of actions $u_{t:t+C-1}$ are predicted and executed in
450 open-loop. Let $u_t \in \mathbb{R}^d$ denote a primitive robot action and let C denote the chunk horizon. We
451 define the action space used by the policy as $\mathcal{A} \subseteq \mathbb{R}^{C \times d}$, so that each action $a_t \in \mathcal{A}$ corresponds to
452 an entire action chunk, $a_t = (u_t, \dots, u_{t+C-1})$. After sampling $a_t \sim \pi(\cdot \mid s_t)$, the robot executes
453 the chunk open-loop for C environment steps before replanning from the resulting state s_{t+C} . For
454 notational simplicity, we treat each chunk as a single action in the MDP, preserving the standard
455 MDP notation above.

456 When using chunked actions, the reward associated with a transition is the discounted reward ac-
457 cumulated over the executed chunk, and bootstrapping occurs from s_{t+C} with discount γ^C . Thus,
458 unless stated otherwise, a_t denotes an action chunk and r_t denotes the reward accumulated over the
459 executed chunk, with bootstrapping discount γ^C .

460 B Related Work

461 C Residual Policy Parameterization

462 D Simulation Environment Details

463 D.1 Libero and Robocasa

464 D.2 Libero Pro Perturbation

465 D.3 Libero Pro State Shift Perturbation

466 E Implementation Details

467 E.1 PARL

468 E.2 BC Implementation

469 E.3 Success Buffer

470 F Ablations

471 **The action head seeds task-relevant experience.** Does the action head need to remain part of
472 the deployed policy, or is it mainly useful for generating task-relevant experience before online
473 learning begins? We test this by removing the base action from the online policy. In this variant,
474 π_{base}^K generates rollouts only during the offline warmup phase; during online adaptation, the learned
475 policy executes actions from VLM features alone, without conditioning on the base-policy action.
476 This variant nearly matches the default MiDAS recipe (Table ??). This suggests that the action
477 head’s main contribution in MiDAS is to seed task-relevant experience, rather than to provide a
478 control signal used by the adapted policy. Once warmup rollouts place the learner in task-relevant
479 regions of the state space, pretrained VLA features and online value learning recover performance.

Task	BC Success Rate (%)		
	$K=1$	$K=3$	$K=5$
<i>LIBERO-Long</i>			
Alphabet Soup + Cream Cheese → Basket	12.7	39.3	82.7
Alphabet Soup + Tomato Sauce → Basket	4.0	32.7	74.7
Black Bowl → Bottom Drawer	22.7	90.0	94.0
Book → Caddy	37.3	26.7	89.3
Both Moka Pots → Stove	13.3	51.3	35.3
Cream Cheese + Butter → Basket	34.0	90.7	84.7
Moka Pot → Stove	84.0	86.0	94.7
White Mug + Chocolate Pudding	33.3	26.7	66.0
White Mug + Plates	30.0	48.7	74.0
Yellow-White Mug → Microwave	64.0	70.7	86.0
Average	33.5	56.3	78.1

Table 2: Per-task BC success on LIBERO-Long. Bold denotes the best result per task across $K \in \{1, 3, 5\}$ demonstrations.

G Curriculum Enables Stronger Position Generalization

G.1 Tables and Full plots

H Position Generalization Analysis

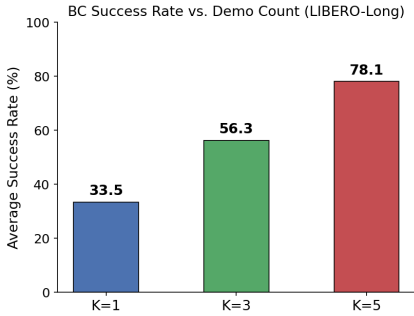


Figure 9: BC success scales with demonstration count. Average BC success on LIBERO-Long at $K=1, 3, 5$. Per-task breakdown in Table 2. Success for $K=0$ is zero across.

Position perturbations expose a gap between ID success and state shift robustness. We isolate this effect using position perturbations at different magnitudes. We compare π_{RL}^1 to π_{base}^1 and to π_{base}^{50} , a behavior-cloned policy trained on all 50 human-teleoperated demonstrations for the task. Although π_{RL}^1 and π_{base}^{50} achieve similar in-distribution success, Fig. 10b shows that π_{base}^{50} generalizes much better as perturbations grow. *The gap is not explained by the residual parameterization alone.* One possibility is that π_{RL}^1 generalizes poorly because the residual actor is less expressive than the pretrained policy architecture. To test this, we distill 50 successful rollouts from π_{RL}^1 back into the base policy, yielding $\pi_{\text{distill}}^{50}$. This still does not recover the robustness of π_{base}^{50} ; in fact, $\pi_{\text{distill}}^{50}$ performs worse than π_{RL}^1 under position perturbations. This suggests that the gap is driven not by policy parameterization, but by differences in state coverage and motion priors between human expert demonstrations and autonomous data. We visualize this coverage in

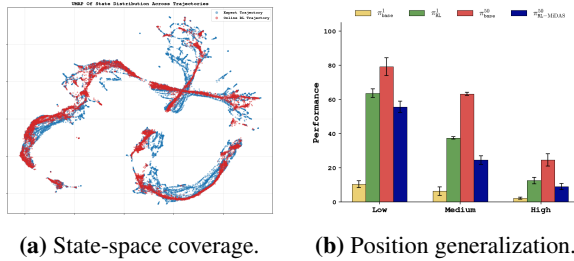


Figure 10: (a) π_{RL}^1 covers a narrow region of state space around its warm-start demo; π_{base}^{50} visits a much broader distribution. **(b)** Success degrades with displacement magnitude; distilling π_{RL}^1 rollouts into the base policy does not close the gap with π_{base}^{50} .

Fig. 10a, where the 50 human demonstrations span a broader region of state space than successful autonomous trajectories, further supporting this explanation.

I Baselines

I.1 Flow matching policy BC baseline

J Real World Implementation Details

K Future Work

[sv: We need to add like demos matter – not all demos are equal]

Table 3: Success rate (%) of plain image-BC flow-matching policies on LIBERO-Long. Final checkpoint, 50 canonical init states.

Task	SR (%)
<i>LIBERO-Long</i>	
Alph. Soup + Cr. Cheese → Basket	0.0
Alph. Soup + Tom. Sauce → Basket	0.0
Black Bowl → Bottom Drawer	28.0
Book → Caddy	0.0
Both Moka Pots → Stove	0.0
Cr. Cheese + Butter → Basket	0.0
Moka Pot → Stove	30.0
White Mug + Choc. Pudding	0.0
White Mug + Plates	0.0
Yellow-White Mug → Microwave	50.0
Average	10.8